

Hi Nusrat,

Question 1

A telecommunications company wants to analyze customer feedback collected from service reviews.

You are given an Excel file containing customer review data about internet and mobile services.

Using Python and NLTK, perform the following tasks:

1. Load the dataset.
2. Clean and normalize the review text.
3. Tokenize the reviews into words.
4. Remove English stopwords and punctuation.
5. Create a new column containing the cleaned version of each review.
6. Calculate the frequency of the most common words across all reviews.
7. Create a new column called `review_length` showing the number of words in each cleaned review.
8. Based on the presence of positive and negative keywords, create a simple new column called `sentiment_label` with values such as Positive, Negative, or Neutral.
9. Display the first 10 processed rows and briefly interpret the results.

Question 2

A company wants to analyze communication patterns among its franchise branches. You are given an Excel file that contains communication relationships between franchise locations. Each row shows:

- the franchise branch sending the communication,
- the franchise branch receiving the communication,
- and the number of communication exchanges between them.

Using Python, pandas, NetworkX, and PyVis, answer the following:

1. Load the Excel file into Python.
2. Create a **directed weighted graph** from the dataset.
3. Print the list of nodes and edges.
4. Report the total number of nodes and edges in the network.
5. Compute the **in-degree** and **out-degree** of each franchise branch.
6. Compute the **degree centrality** and **betweenness centrality** of each franchise branch.
7. Identify:
 - the most connected franchise branch,
 - and the most important bridge franchise branch in the network.
8. Find the **shortest path** from Franchise_A to Franchise_L.
9. Create a **basic network visualization** using NetworkX.
10. Create an **interactive visualization** using PyVis, where node size reflects degree centrality and edge labels show communication counts.

Question 3

A company wants to analyze job applicants, interview results, and department hiring needs.

You are given three datasets:

1. **applicants**: applicant information
2. **interviews**: interview records and scores
3. **departments**: department hiring information

Using these datasets, do the following:

- Inspect the structure, columns, and shapes of all datasets.
- Identify the common key columns for combining the tables.
- Merge the datasets into one final business table.
- Detect unmatched records across the tables.
- Check whether any key columns contain duplicated values and explain how they affect the merge.
- Identify missing values in the final merged dataset.
- Create a new column that classifies interview scores into **Strong**, **Average**, and **Weak**.
- Create a pivot table showing the **average interview score by department**.
- Create another summary showing the **number of applicants by application status**.
- Reshape the data if needed to improve the comparison of interview stages.
- Write a short interpretation of the business insights from the final results.